Toktam Samiei

Computational Neuroscience · Neural Population Coding · NeuroAI

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Research Profile

I am a Ph.D. candidate at the University of California, Riverside, specializing in computational neuroscience and machine learning, with a focus on neural population coding. My research uses neural decoding to investigate how the brain represents and processes information, while examining how the statistical and dynamical properties of population activity shape what can be decoded and at which spatiotemporal scales. I combine large-scale electrophysiological recordings with mathematical modeling and develop decoding and optimization frameworks to address these questions. I also use Transformer architectures to investigate the temporal organization of neural activity and information contributed by interactions within and across brain regions. My work has identified mesoscale optimality in neural decoding and distinct forms of temporal organization in sensory population activity.

Research Contributions and Ongoing Projects

Graduate Research Assistant, University of California, Riverside September 2022–Present

Advisor: Prof. Erfan Nozari

At What Spatial and Temporal Resolutions Is Neural Information Best Decoded?

Found that visual information is most accurately decoded at mesoscale spatial and temporal resolutions, rather than at the finest or coarsest scales. Optimal temporal resolution varied across brain regions and correlated with theta-band oscillations; spatial and temporal averaging also showed an interdependence. Developed NeuroPixelHD, a symbolic hyperdimensional encoding–decoding model of multi-unit activity designed to control spatiotemporal averaging.**(Frontiers in Cellular Neuroscience, 2024; first author)[GitHub Repository]** In a complementary theoretical study, contributed to establishing how the temporal autocorrelation of signal and noise determines when integration improves stimulus discriminability and when excessive averaging degrades it. **(Frontiers in Computational Neuroscience, 2026; second author)**

How Do Rate and Temporal Coding Coexist in Visual Population Activity?

Identified *bracket coding*, in which population spiking activity is rate-coded within temporally bounded intervals separated by precisely timed, population-synchronized boundaries. Developed a sweeping-integration analysis to identify boundaries from changes in stimulus-decoding error; integrating activity within the identified brackets improved decoding relative to random temporal partitions. Bracket coding strengthened along the thalamocortical visual hierarchy (LGd–V1–HVAs), showed predominantly bottom-up long-range synchrony, and was phase-locked to low-frequency LFP oscillations. Core findings were reproduced in International Brain Laboratory recordings, while simulations showed that brackets can emerge from synchronized changes in firing-rate profiles. **(bioRxiv, 2026; first-author preprint)[GitHub Repository]**

Is the Decoding-Optimal Temporal Scale 'Dynamic'? and What Explains Its Evolution?

Found that the optimal temporal scale contracts after stimulus onset, reaches its finest resolution near peak stimulus decodability, and subsequently re-expands within a single sensory response, across four visual paradigms. Developed Integration–Split, a decoding-guided optimization framework that adaptively merges and splits temporal intervals. The window maximizing an autocorrelation-based signal-to-noise measure closely tracked the optimal-scale trajectory, while nonlinear predictive advantage increased as the scale became finer. A spike-event Transformer trained only to decode stimulus identity learned similar temporal organization without timescale supervision; attention-head ablations supported the functional relevance of this learned structure. These associations do not establish a causal mechanism. **(bioRxiv, 2026; first-author preprint) [GitHub Repository]**

Do Intrinsic Neuronal Timescales Predict the Optimal Decoding Resolution?

Investigating whether intrinsic timescales estimated from single-neuron spike-train autocorrelation predict the temporal window that maximizes population decoding. Developing controlled excitatory/inhibitory spiking-network simulations to examine the effects of neuronal recovery, synaptic decay, transmission delays, and coupling on intrinsic activity timescales versus decoding-optimal temporal resolution. *(Ongoing research.)*

Do Cross-Region Interactions Provide Synergistic Information for Neural Decoding?

Investigated whether simultaneous recordings from different brain regions provide information beyond their individual contributions to stimulus and choice decoding. Developed region-specific Transformer encoders and cross-attention architectures, compared with linear fusion, logistic-regression baselines, and partial information decomposition. Combining visual regions improved stimulus decoding, but explicit cross-attention did not consistently outperform fusion of separately learned regional representations. Pre-movement choice decoding showed similarly limited gains, suggesting that much of the benefit in these datasets reflects complementary information rather than strong interaction-dependent synergy. *(Ongoing research.)*

Can Directed Neuronal Interactions Be Inferred from Spike Trains?

Developing an attention-based Transformer for spike-train self-prediction and candidate connectivity discovery. The architecture combines temporal self-attention with inter-neuronal cross-attention and is being evaluated against known connectivity in simulated excitatory/inhibitory spiking networks. Predictive ablations test the relevance of inferred connections without treating attention weights alone as evidence of causality. *(Ongoing research.)*

Research Assistant, University of Tehran September 2021–April 2022

Studied multimodal emotion recognition from EEG and eye-movement recordings using deep canonical correlation analysis to learn shared representations.

Research Intern, University of Tehran June–September 2020

Investigated ultrasound-induced refractive-index modulation for guiding light through scattering tissue, with applications to non-invasive neurotechnology.

Education

Ph.D., Mechanical Engineering, University of California, Riverside Expected June 2027

Advisor: Prof. Erfan Nozari; research focus: computational neuroscience, neural decoding, and machine learning.

M.S., Mechanical Engineering, University of California, Riverside June 2024

Advisor: Prof. Erfan Nozari.

B.S., Engineering Science, University of Tehran April 2021

Electrical Engineering area of study; ranked 2nd.

Presentations, Teaching, and Academic Service

Conferences: Society for Neuroscience (SfN) Annual Meeting: posters presented in 2023, 2024 and 2025; poster accepted for 2026. Selected to participate in the Lake Conference on Neural Coding and Dynamics.
Teaching: Linear Control Systems (four offerings); Dynamical Systems; Electromagnetics; Physics II (two offerings).
Peer review: Ad hoc reviewer for *Nature Methods*, *Nature Communications*, *Communications Physics*, *Chaos*, and *iScience*.
Training: Neuromatch Academy, Computational Neuroscience Summer School (2022).

Selected Technical Skills

Programming and scientific computing: Python, MATLAB, C++; PyTorch, TensorFlow/Keras, snnTorch, NumPy/SciPy, scikit-learn; GPU/HPC computing, SLURM.
Machine learning and modeling: Transformers, representation learning, recurrent and spiking neural networks, hyperdimensional computing, dynamical systems, optimization, statistical and nonlinear predictive modeling.
Neural data analysis: Neuropixels, LFP, EEG/iEEG, calcium imaging, eye tracking; neural decoding, signal processing, information theory, statistical inference; Allen Brain Observatory and IBL datasets.

Selected coursework:

Artifitial Inteligence, Machine Learning; Deep Learning; Reinforcement Learning, COmputer Vision, Digital Signal Processing; Linear Algebra; Dynamical Systems; Optimal Control and Estimation; Cellular Neuroscience.